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G H Raisoni International Skill Tech University, Pune

F.Y. B. Tech

TERM-I (Regular)

CAE I- WINTER 2025 (2024 Pattern)

Indian Knowledge System (25UHSL1202)

(Time: - 01 Hour) **(Max. Marks: 20)**

Instructions to the students:

i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	*Question*	Marks	CO	BL
Q1.(a)	Define vinculum numbers. Convert the number 9768 and 16789 into its vinculum form using the Vedic approach with each step.	[5]	1	3
b)	Find Cube of: a) 25 b) 33 using the Vedic approach with each step	[5]	1	2
	OR			
Q 2.(a)	Convert Vinculum to Normal Number 1) $4\bar{2}\bar{5}\bar{1}1$ 2) $2\bar{1}\bar{3}4\bar{3}1$ 3) $3\bar{1}\bar{3}4\bar{5}\bar{3}$	[5]	1	3
(b)	Convert $a)\ (123)_4 = (?)_{10}$ $b)\ (84)_6 = (?)_{10}$ $c)\ (213)_5 = (?)_{10}$	[5]	1	2
Q 3.(a)	Check whether the following numbers a) 1234560 is divisible by 8 b) 2002 is divisible by 7 c) 23564 is divisible by 4	[5]	2	3
(b)	Find HCF of a) 48, 64, 80 b) 119, 272	[5]	2	2
	OR			
Q. 4 (a)	Find LCM of a) 12, 16, 20 b) 18, 24, 32 c) 72, 108	[5]	2	2
(b)	Check whether the following numbers a) 153450 is divisible by 5 b) 20879 is divisible by 3 c) 219492 is divisible by 6	[5]	2	3